

a: Solve the 8 Queens Problem

Aim:

To place 8 queens on an 8x8 chessboard so that no two queens attack each other, i.e., no two queens share the same row, column, or diagonal.

Procedure:

1. Create an 8x8 board initialized with 0s.
2. Place a queen in each row by checking if it's safe.
3. If no column is safe in a row, backtrack.
4. Repeat until all 8 queens are placed.

Code:

```
N = 8
solution_count = 0
def is_safe(board, row, col):
    for i in range(row):
        if board[i][col] == 1:
            return False
    i, j = row - 1, col - 1
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1
    i, j = row - 1, col + 1
    while i >= 0 and j < N:
        if board[i][j] == 1:
            return False
        i -= 1
        j += 1
    return True
def print_board(board):
    for row in board:
        for cell in row:
            if cell == 1:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
    print()
def solve(board, row):
    global solution_count
```

```

if row == N:
    solution_count += 1
    print(f"Solution {solution_count}:")
    print_board(board)
    return
for col in range(N):
    if is_safe(board, row, col):
        board[row][col] = 1
        solve(board, row + 1)
        board[row][col] = 0
board = [[0 for _ in range(N)] for _ in range(N)]

solve(board, 0)

print("Total Solutions =", solution_count)

```

Output:

Solution 1:

```

Q .....
.... Q ...
..... Q
..... Q ..
.. Q .....
..... Q .
. Q .....
... Q .....

```

Solution 2:

```

Q .....
..... Q ..
..... Q
.. Q .....
..... Q .
... Q .....
. Q .....
..... Q ...

```

Solution 3:

```

Q .....
..... Q .
... Q .....
..... Q ..
..... Q
. Q .....

```

```
....Q...  
..Q.....
```

Solution 4:

```
Q.....  
.....Q.  
....Q...  
.....Q  
.Q.....  
...Q....  
.....Q..  
..Q.....
```

Result:

Thus the program of placing 8 queens on an 8x8 chessboard was executed successfully.